

Evaluating a Classifier

Rest of our Schedule

Lecture Course-2026 Spring			
	ET Time (12 hrs time difference)	China Time	
Lecture 1	6-8AM on Saturday, March 14	6-8PM on Saturday, March 14	
Lecture 2	6-8:20AM on Saturday, March 21	6-8:20PM on Saturday, March 21	
Lecture 3	6-8:20AM on Saturday, March 28	6-8:20PM on Saturday, March 28	Homework 1
Lecture 4	6-8AM on Saturday, April 4	6-8PM on Saturday, April 4	Homework 2
	6-8AM on Saturday, April 11	6-8PM on Saturday, April 11	
Lecture 5	6-8 AM Friday, April 17	6-8pm Friday April 17	
Mid-term (Specific Time TBD, on the weekend of April 18/19)			
Lecture 6	6-8AM on Saturday, April 25	6-8PM on Saturday, April 25	
Lecture 7	6-8AM on Saturday, May 2	6-8PM on Saturday, May 2	
Lecture 8	6-8AM on Saturday, May 9	6-8PM on Saturday, May 9	Homework 3
Lecture 9	6-8AM on Sunday, May 10	6-8PM on Sunday, May 10	
	6-8AM on Saturday, May 16	6-8PM on Saturday, May 16	Homework 4
Lecture 10	3:30 pm - 5:30 pm on Saturday, May 23 (Raja from India)	6-8PM on Saturday, May 23	
Final Term (Specific Time TBD, on the weekend of May 30/31)			

Saturday, midterm exam

- 20-30 multiple choice questions
- please review all of the slides from the first 5 classes
 - 01-0308-course-intro-neo-s25.pdf
 - 02-0316-computational-thinking-tidy-data-neo-s25.pdf
 - 03-0322-visualization-neo-s25.pdf
 - 04-ml-intro-neo-s25.pdf
probability-review-s25.pdf
 - 05-0405-classifier-evaluation-neo-s25.pdf

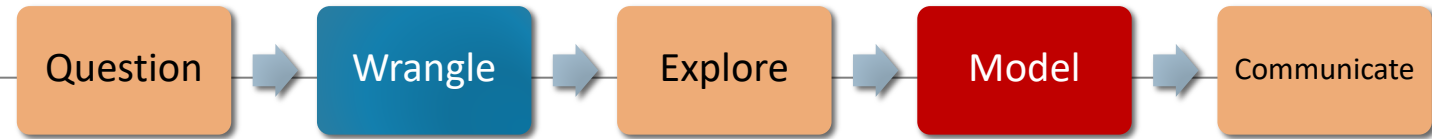
Learning Quantified

- A Computer Program is said to learn from **experience E** with respect to some **class of tasks T** and **performance measure P**, if its performance at tasks in T, as measured by P, improves with E

... Tom Mitchell

$$P(T, E+\text{more}) > P(T, E)$$

ML: Task, Technique, Tool



- **Classification / Regression**

- assign a label (classification) or numerical value (regression) to an unknown entity based on a set of features and known labels (or numerical values)

- **Clustering**

- group a bunch of entities that share common features

- **Optimization**

- from amongst a set of alternatives pick the “best” while balancing competing value metrics

- **Forecasting**

- based on the past, forecast the future

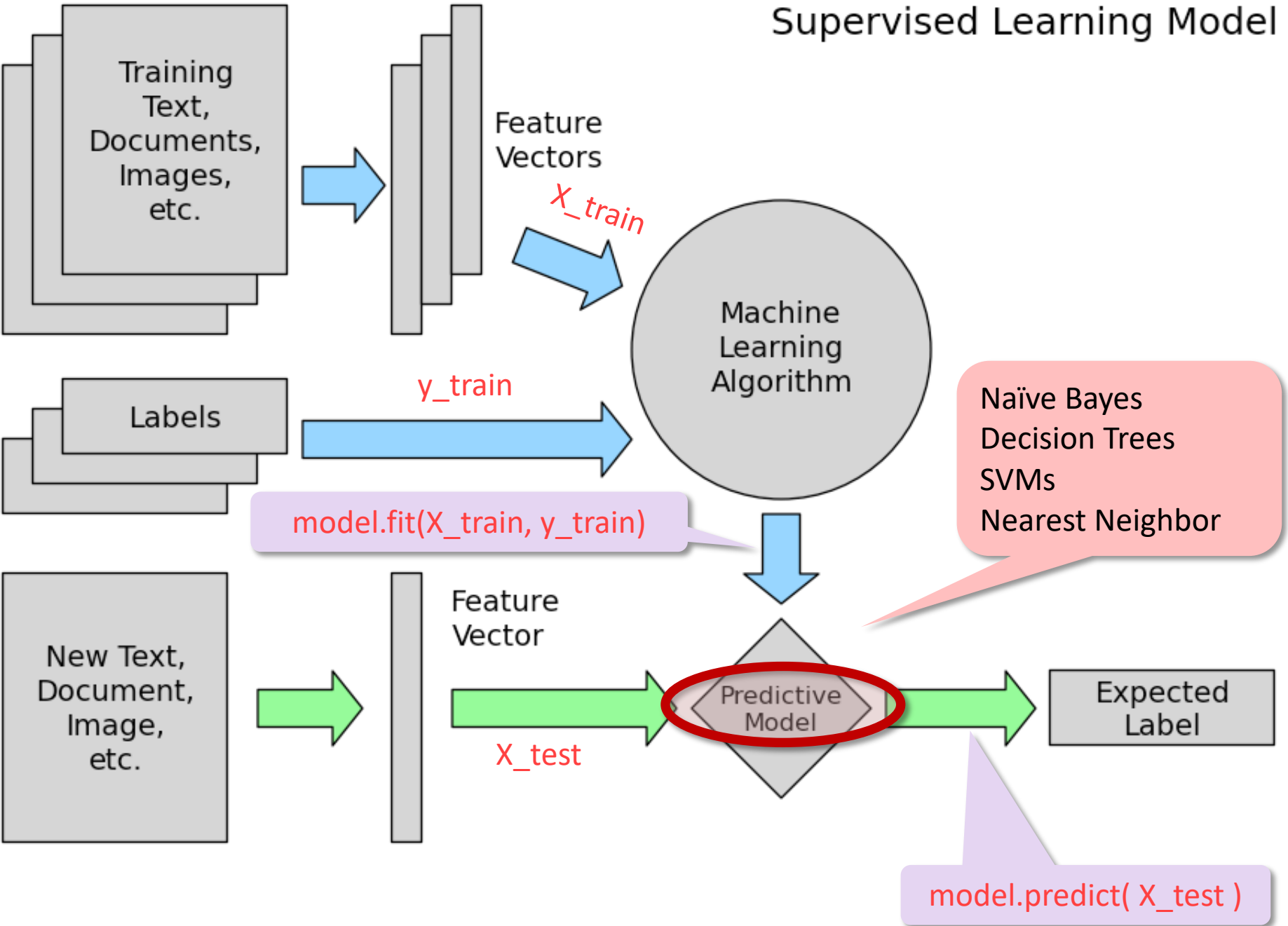
- **Recommendation**

- based on prior behavior rank order candidate preferences

- **Association**

- identify which items co-occur e.g., bread and peanut butter

Supervised Learning Model

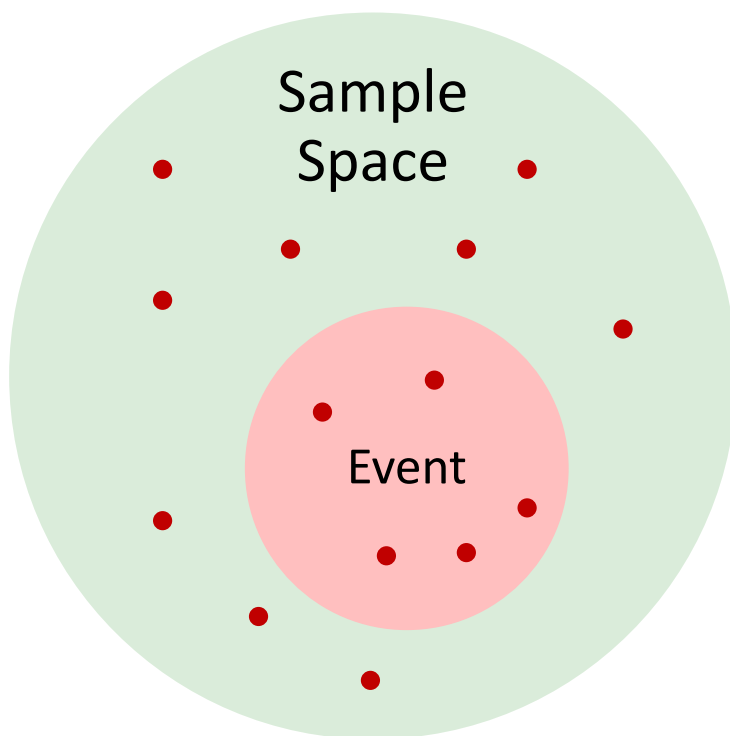


When the facts change,
I change my opinion.
What do you do, sir?
—John Maynard Keynes


$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

HOW A NB CLASSIFIER WORKS

An Event is a subset of the Sample Space



PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Event: (Outlook=Sunny)
{ D1, D2, D8, D9, D11 }

Classical Probability

$$\text{Probability of event E} = \frac{\text{Number of outcomes in an event E}}{\text{Total number of outcomes in Sample Space}}$$

Outcome: result of an experiment

Event: set of outcomes (subset of sample space)

Assumption: all outcomes are equally likely

$$P(\text{sunny}) = 5/14$$

Calculating and Estimating Probabilities

1. Counting

$$\frac{\text{\# of Outcomes in event E}}{\text{\# of Outcomes in Sample Space}}$$

2. Exact Bayes' rule

$$P(C | f_1, f_2) = \frac{P(f_1, f_2 | C) * P(C)}{P(f_1, f_2)}$$

3. Naïve (approximate) Bayes' rule

- using the independence assumption

$$P(C | f_1, f_2) = \frac{P(f_1 | C) * P(f_2 | C) * P(C)}{P(f_1, f_2)}$$

-

$$P(C_1 | f_1 f_2) = \frac{P(f_1 | C_1) * P(f_2 | C_1) * P(C_1)}{P(f_1 | C_1) * P(f_2 | C_1) * P(C_1) + P(f_1 | C_2) * P(f_2 | C_2) * P(C_2)}$$

4. Relative Estimation

$$P(f_1 | C_1) * P(f_2 | C_1) * P(C_1)$$

Back to our example ...

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
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D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)$

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) = \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

Focusing on the numerators ... we get ...

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

$$P(\text{no} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx \frac{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{no}) * P(\text{no})}{P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle)}$$

Exercise: Compute and Compare (Green handout)

$$P(\text{yes} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{yes}) * P(\text{yes})$$

$$P(\text{Yes} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{Yes}) P(\text{Cool} \mid \text{Yes}) P(\text{High} \mid \text{Yes}) P(\text{Strong} \mid \text{Yes})] * P(\text{Yes})$$

$$P(\text{no} \mid \langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle) \approx P(\langle \text{Sunny}, \text{Cool}, \text{High}, \text{Strong} \rangle \mid \text{no}) * P(\text{no})$$

$$P(\text{No} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{No}) P(\text{Cool} \mid \text{No}) P(\text{High} \mid \text{No}) P(\text{Strong} \mid \text{No})] * P(\text{No})$$

1. Compute the tables for each feature (Outlook, Temperature, Humidity, Wind)
2. Compute the probability of each category (yes, no)
3. Compute the individual probabilities

Complete these tables (Green handout)

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot		
Mild		
Cool		

Humidity	Play=Yes	Play=No
High		
Normal		

Wind	Play=Yes	Play=No
Strong		
Weak		

$$P(\text{Play=Yes}) = 9/14$$

$$P(\text{Play=No}) = 5/14$$

From these tables

Outlook	Play=Yes	Play=No
<i>Sunny</i>	2/9	3/5
<i>Overcast</i>	4/9	0/5
<i>Rain</i>	3/9	2/5

Temperature	Play=Yes	Play=No
<i>Hot</i>	2/9	2/5
<i>Mild</i>	4/9	2/5
<i>Cool</i>	3/9	1/5

Humidity	Play=Yes	Play=No
<i>High</i>	3/9	4/5
<i>Normal</i>	6/9	1/5

Wind	Play=Yes	Play=No
<i>Strong</i>	3/9	3/5
<i>Weak</i>	6/9	2/5

$$P(\text{Play=Yes}) = 9/14$$

$$P(\text{Play=No}) = 5/14$$

Relative Estimation

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=Yes	Play=No
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=Yes	Play=No
Strong	3/9	3/5
Weak	6/9	2/5

$\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

Look up from tables

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Play}=\text{No}) = 5/14$$

$$P(\text{Play}=\text{Yes}) = 9/14 \quad P(\text{Play}=\text{No}) = 5/14$$

Relative Estimation

Outlook	Play=Yes	Play=No
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=Yes	Play=No
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=Yes	Play=No
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=Yes	Play=No
Strong	3/9	3/5
Weak	6/9	2/5

$\mathbf{x}' = (\text{Outlook}=\text{Sunny}, \text{Temperature}=\text{Cool}, \text{Humidity}=\text{High}, \text{Wind}=\text{Strong})$

Look up from tables

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{Yes}) = 2/9$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{Yes}) = 3/9$$

$$P(\text{Play}=\text{Yes}) = 9/14$$

$$P(\text{Outlook}=\text{Sunny} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Temperature}=\text{Cool} \mid \text{Play}=\text{No}) = 1/5$$

$$P(\text{Humidity}=\text{High} \mid \text{Play}=\text{No}) = 4/5$$

$$P(\text{Wind}=\text{Strong} \mid \text{Play}=\text{No}) = 3/5$$

$$P(\text{Play}=\text{No}) = 5/14$$

$$P(\text{Yes} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{Yes}) P(\text{Cool} \mid \text{Yes}) P(\text{High} \mid \text{Yes}) P(\text{Strong} \mid \text{Yes})] \cdot$$

$$P(\text{Play}=\text{Yes})$$

$$[2/9 \cdot 3/9 \cdot 3/9 \cdot 3/9] \cdot 9/14 = 0.0053$$

$$P(\text{No} \mid \mathbf{x}'): [P(\text{Sunny} \mid \text{No}) P(\text{Cool} \mid \text{No}) P(\text{High} \mid \text{No}) P(\text{Strong} \mid \text{No})] \cdot P(\text{Play}=\text{No})$$

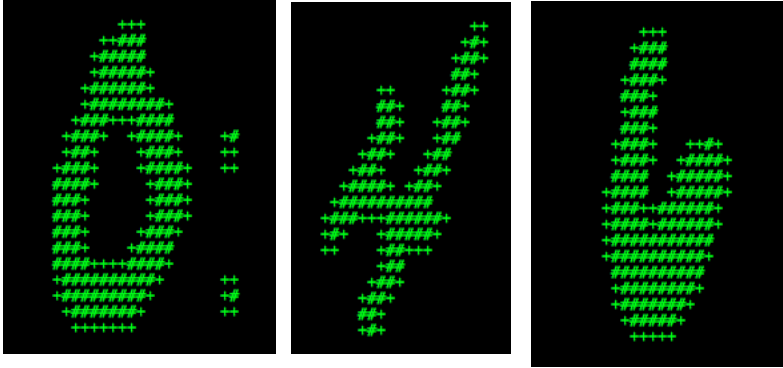
$$3/5 \cdot 1/5 \cdot 4/5 \cdot 3/5 \cdot 5/14 = 0.0206$$

$$P(\text{Play}=\text{Yes}) = 9/14 \quad P(\text{Play}=\text{No}) = 5/14$$

Given $P(\text{Yes} \mid \mathbf{x}') < P(\text{No} \mid \mathbf{x}')$, we classify $\mathbf{x}'$ to be “No”

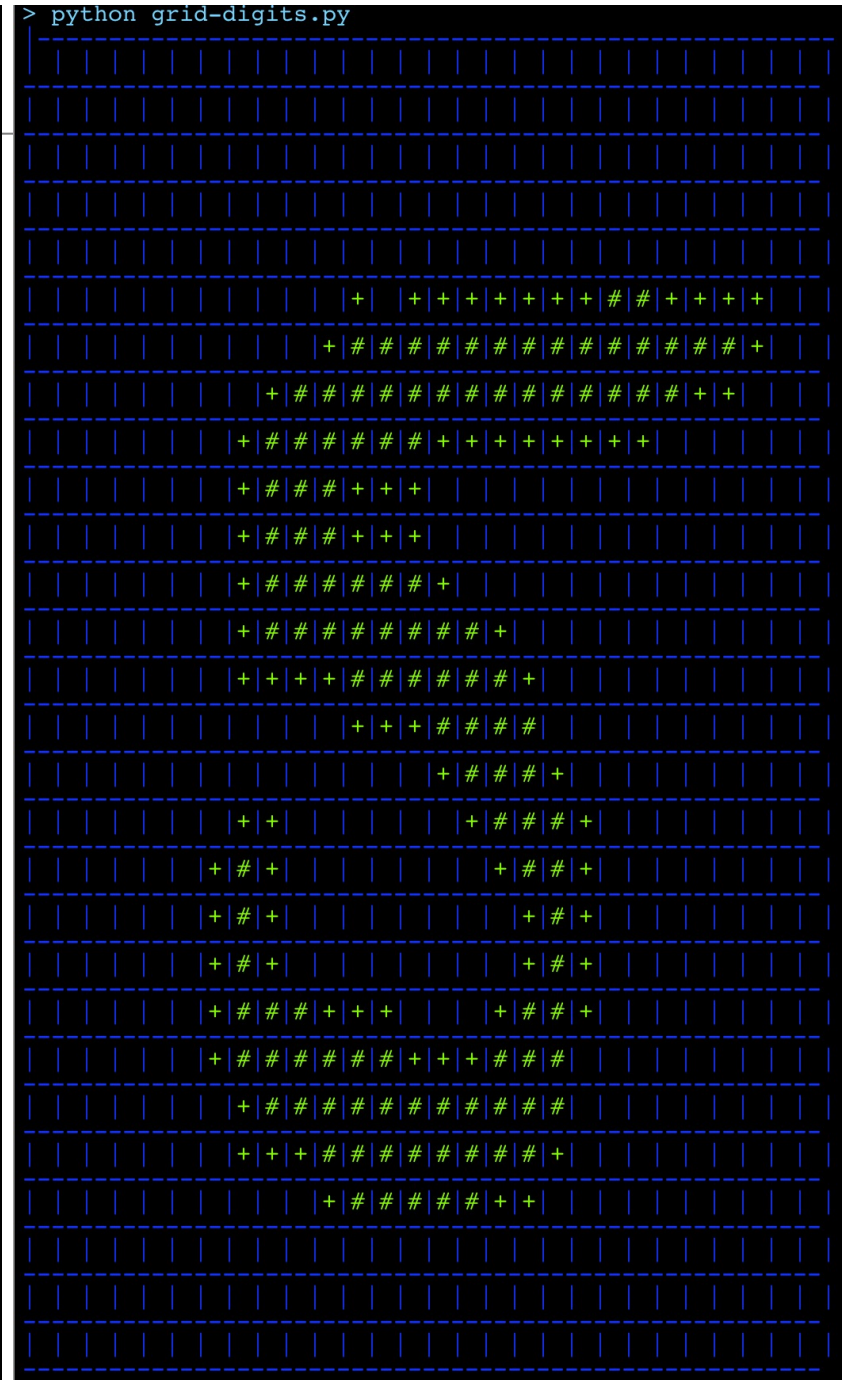
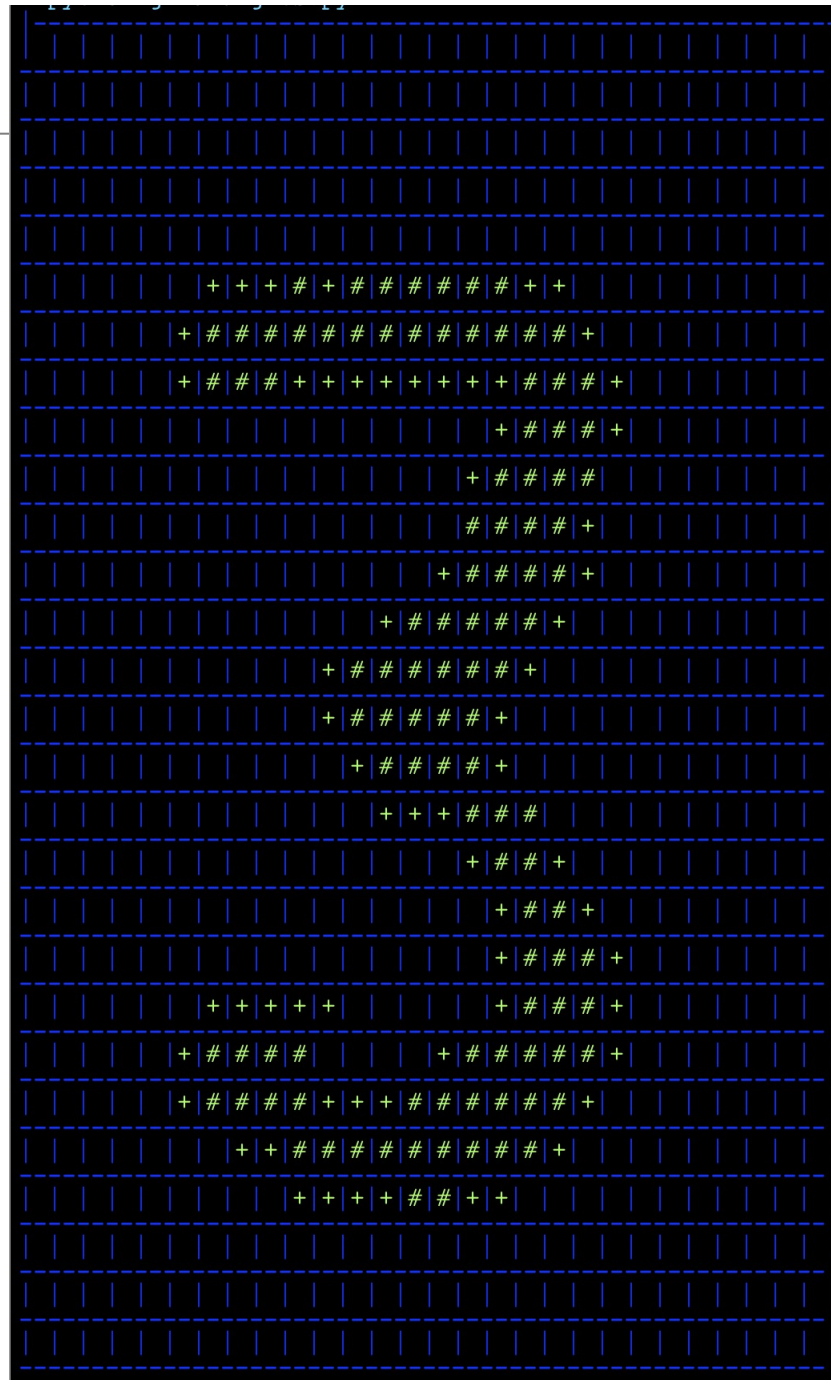
DEMO

MNIST Database

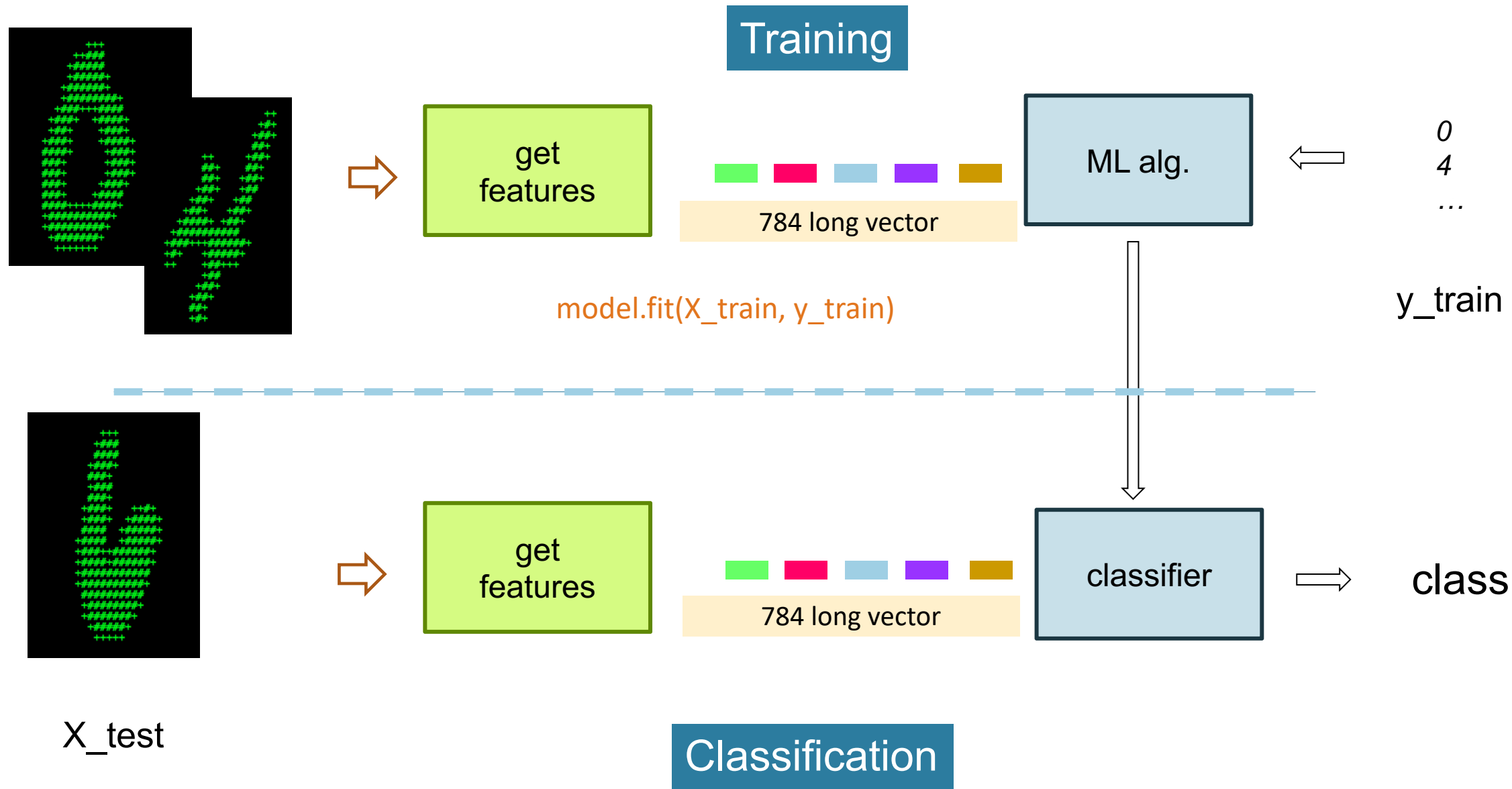


Training: 5000 images
Testing: 1000 images

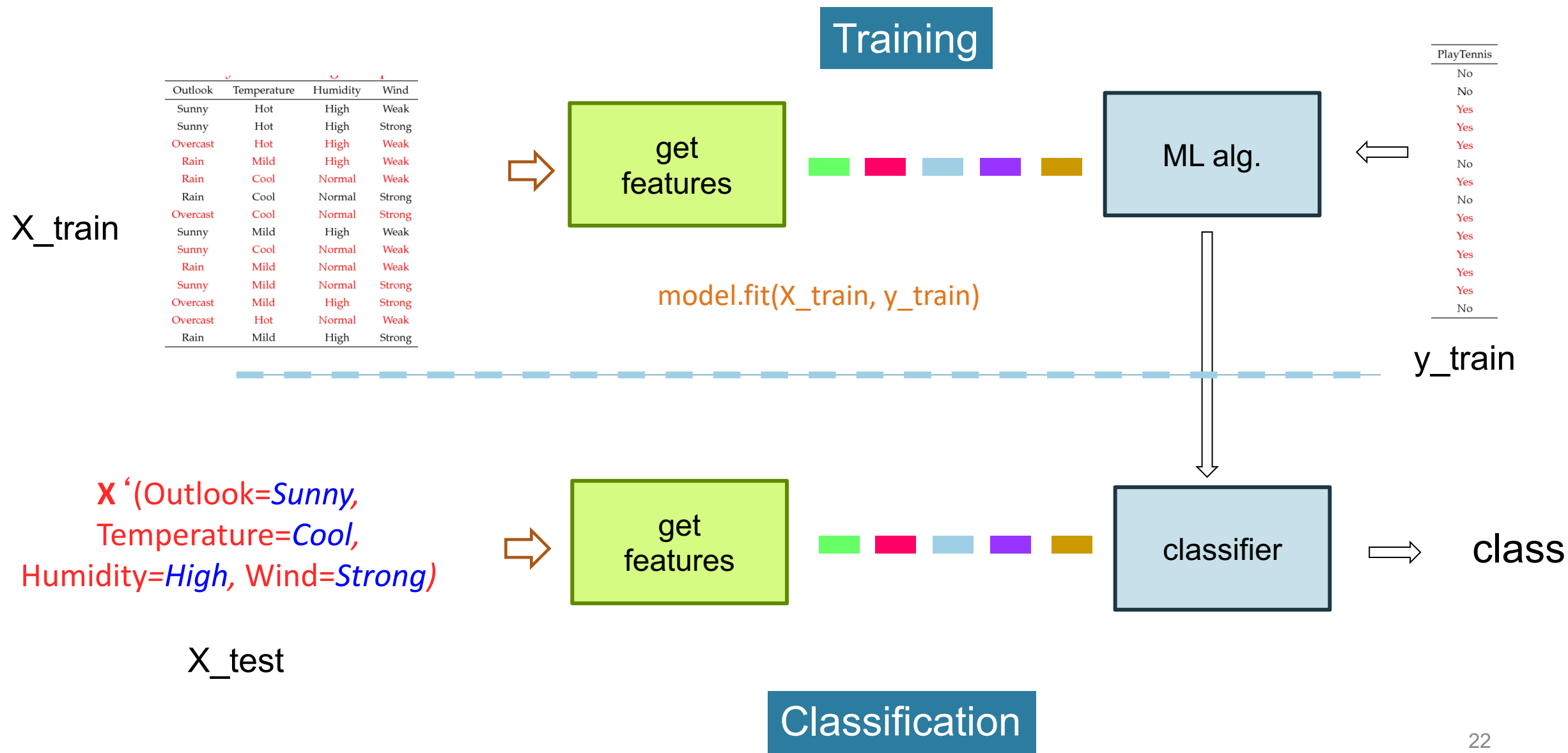
*Each "image" is a
28x28 ascii grid
784 features*



Two Phases: Training + Classification



Two Phases: Training + Classification



CROSS-VALIDATION AND DATA LEAKAGE

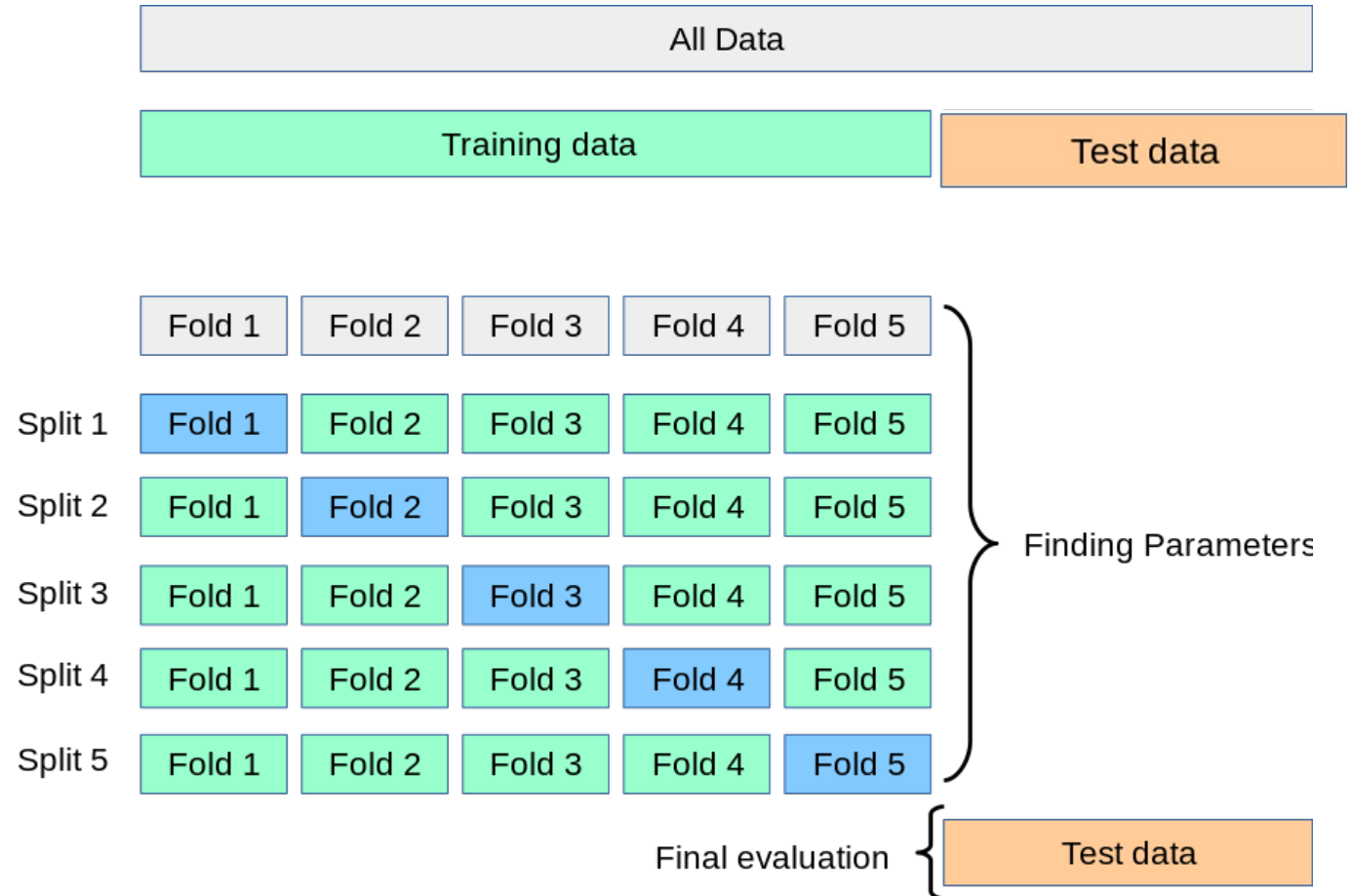
Training data is Training data and

Test data is Test data and never twain shall meet.

(with apologies to Kipling)

Train-Test split vs. k-fold cross-validation

```
X_train, X_test,  
y_train, y_test =  
    train_test_split(X, y,  
                    test_size=0.2,  
                    random_state=1)
```



https://en.wikipedia.org/wiki/Training,_validation,_and_test_data_sets

`sklearn.model_selection.RepeatedStratifiedKFold`

https://scikit-learn.org/stable/modules/cross_validation.html

Leakage Scenario

1. Prepare data set (eliminate redundancy, impute, scale, ...)
2. Split data set
3. Evaluate model

1. Split the data set
2. **Prepare using the training data set**
3. **Apply the same preparation to the Training and Test data sets**
4. Evaluate the model

Data Preparation (80% of the effort)

- Data cleansing
- Feature Selection
- Data Transforms
- Feature engineering
- Dimensionality Reduction

DEMO

01-avoiding-data-leakage

CLASSIFIER PERFORMANCE

Examples

- Learning to play checkers
 - Task T: play checkers
 - Performance measure P: % of games won
 - Training Experience E: playing practice games (with others or itself)
- Learning to recognize handwriting
 - Task T: recognize and classify handwritten words within images
 - Performance measure P: % of words correctly classified
 - Training experience E: a database of handwritten words with pre-specified classifications

Dwarf or Smurf?



Bashful
Brainy
Clumsy
Doc
Dopey
Greedy
Grouchy
Grumpy
Gutsy
Happy
Hefty
Papa
Sleepy
Sneezy

Dwarf or Smurf?: Suppose someone classified as below ...



Bashful

Brainy

Clumsy

Doc

Dopey

Greedy

Grouchy

Grumpy

Gutsy

Happy

Hefty

Papa

Sleepy

Sneezy

Bashful

Brainy

Clumsy

Doc

Dopey

Greedy

Grouchy

Grumpy

Gutsy

Happy

Hefty

Papa

Sleepy

Sneezy

Metrics of a Dwarf detector



Clumsy, Brainy, Grouchy, Papa, Gutsy, Hefty, Greedy

Happy, Dopey, Doc, Grumpy, Sneezy, Bashful, Sleepy



		Predicted	
		0	1
Actual	0	5 TN	2 FP
	1	4 FN	3 TP



Metrics of a Dwarf detector



Clumsy, Brainy, Grouchy, Papa,
Gutsy, Hefty, Greedy

Happy, Dopey, Doc, Grumpy,
Sneezy, Bashful, Sleepy



Type 1 error

		Predicted	
		0	1
Actual	0	5 TN	2 FP
	1	4 FN	3 TP

Type 2 error

- **Accuracy:** Overall, how many did you get right? $(5+3)/(5+2+4+3)$
- **Recall, Sensitivity, TPR:** Of the ones you should have picked, how many did you? $3/(4+3)$
- **Specificity:** Of the ones you should not have picked, how many did you not pick? $5/(5+2)$
- **FPR:** $2/(5+2)$ 1- specificity
- **Precision:** Of the ones you picked, how many were right? $3/(2+3)$

F1 score

- How to combine Precision and Recall into a single value?
- Recall: $TP/(TP+FN)$
- Precision: $TP/(TP+FP)$
- Can't just take the average, since denominators are different. So we take the harmonic mean.
 $34/50 + 56/70$
- $1/((1/R + 1/P)/2) = 2PR/(P+R)$

Excellent article on “averages”:

<https://towardsdatascience.com/on-average-youre-using-the-wrong-average-geometric-harmonic-means-in-data-analysis-2a703e21ea0>

Which error do you want to minimize?

- Spam classifier
- Tumor detection
- Credit card fraud detection
- Credit worthiness

		Predicted	
		0	1
Actual	0	5 TN	2 FP
	1	4 FN	3 TP

Type 1 error

Type 2 error

Which error do you want to minimize?

- Spam classifier
- Tumor detection
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Type 1 error

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Type 2 error

Which error do you want to minimize?

- Spam classifier

- Minimize: FP

- Tumor detection

- Minimize: FN

- Credit card fraud detection

- Minimize: FN

- Credit worthiness

- Minimize: FP

		Predicted	
		0	1
Actual	0	5 TN	2 FP
	1	4 FN	3 TP

Type 1 error

- **Accuracy**: Overall, how many did you get right? $(5+3)/(5+2+4+3)$
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Type 2 error

What a classifier does



A classifier should only report the probability of belonging to a class. The user then determines a threshold above which the user would classify the input as belonging to class-1 or class-0

F1 score

- How to combine Precision and Recall into a single value?
- Recall: $TP/(TP+FN)$
- Precision: $TP/(TP+FP)$
- Can't just take the average, since denominators are different

$34/50 + 56/70$

- $1/((1/R + 1/P)/2) = 2PR/(P+R)$

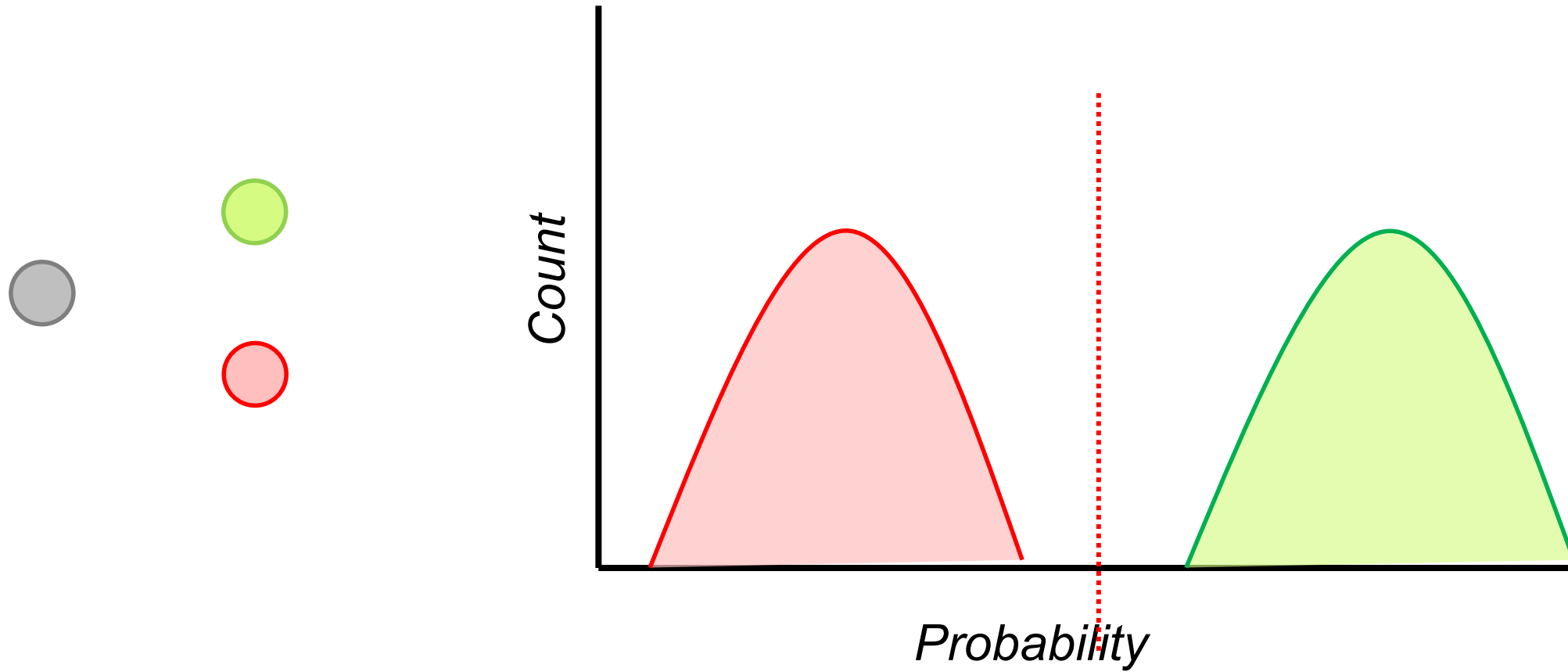
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ROC CURVE AND THE AUC METRIC

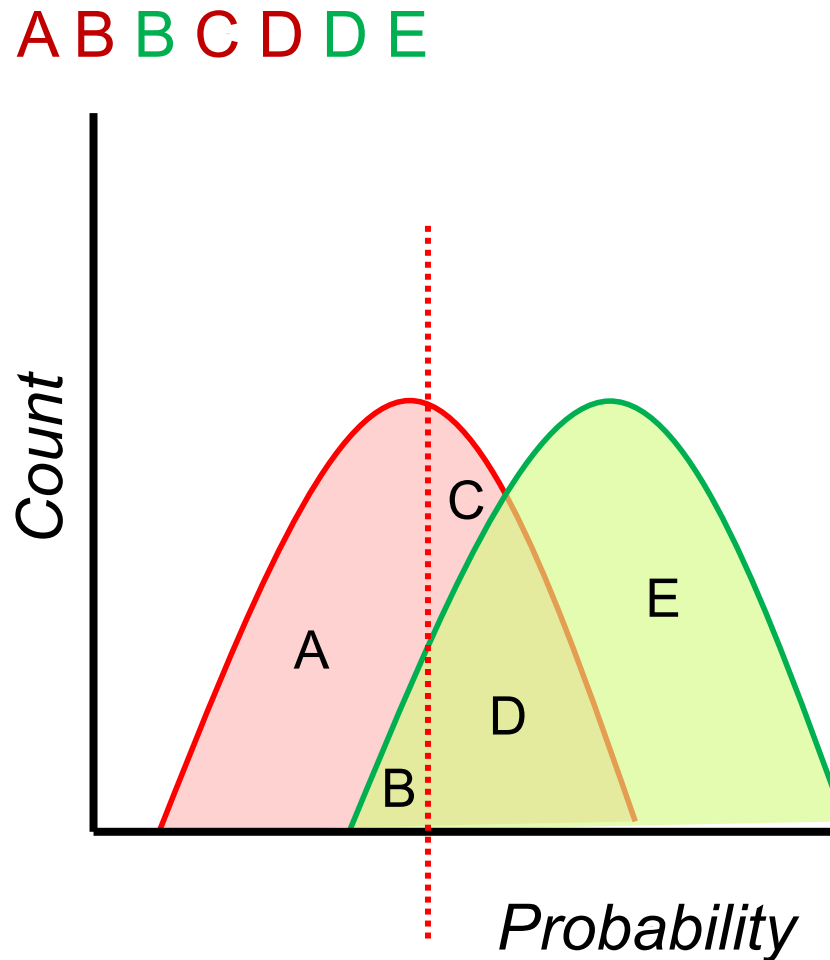
varying the threshold

ROC curves and the classifying thresholds



ROC curves and the classifying thresholds

Define the following concepts in terms of the following 7 non-overlapping areas


$$TP =$$
$$TN =$$
$$FP =$$
$$FN =$$

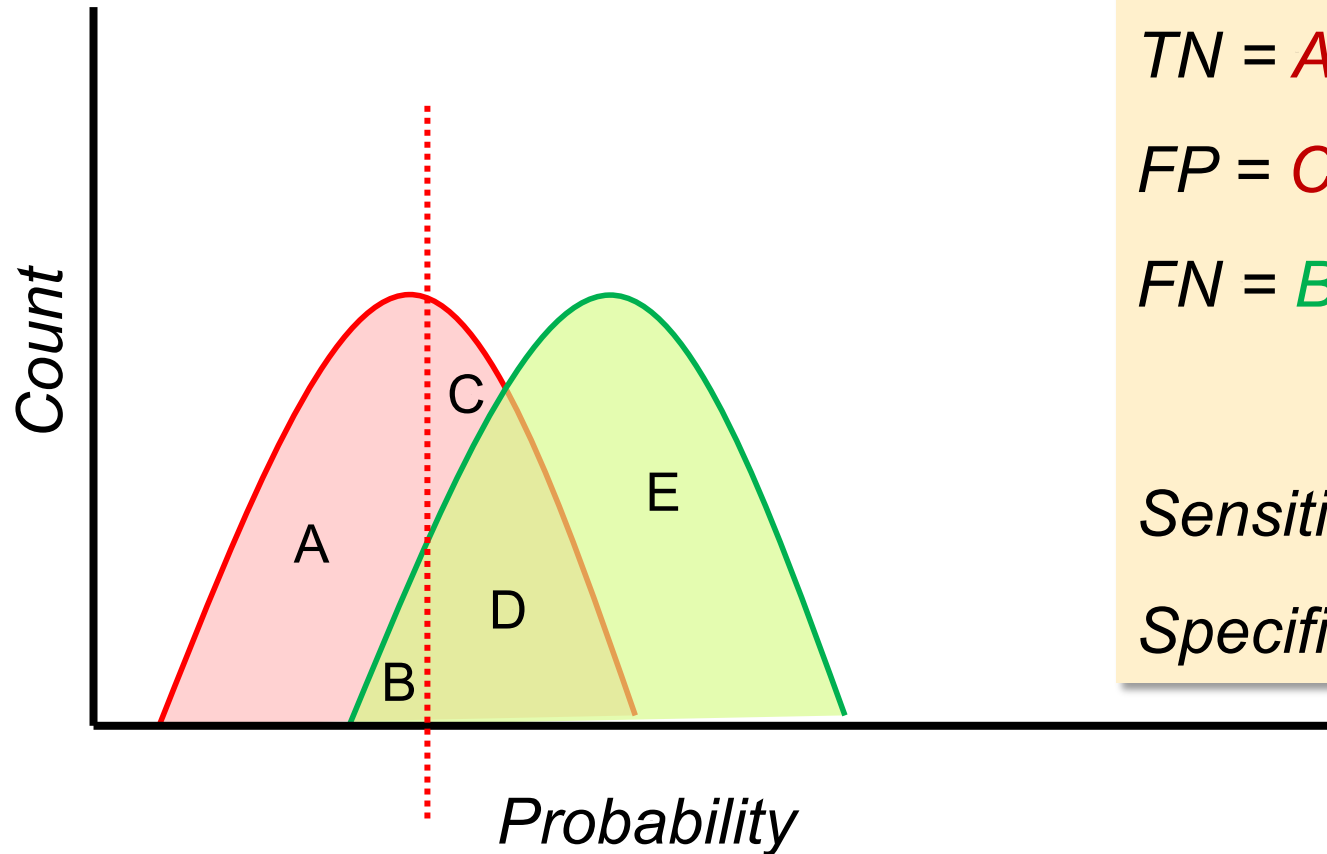
$$Sensitivity = TP/(TP+FN)$$

$$Specificity = TN/(TN+FP)$$

ROC curves and the classifying thresholds

Define the following concepts in terms of the following 7 non-overlapping areas

A B B C D D E



$$TP = E + D$$

$$TN = A + B$$

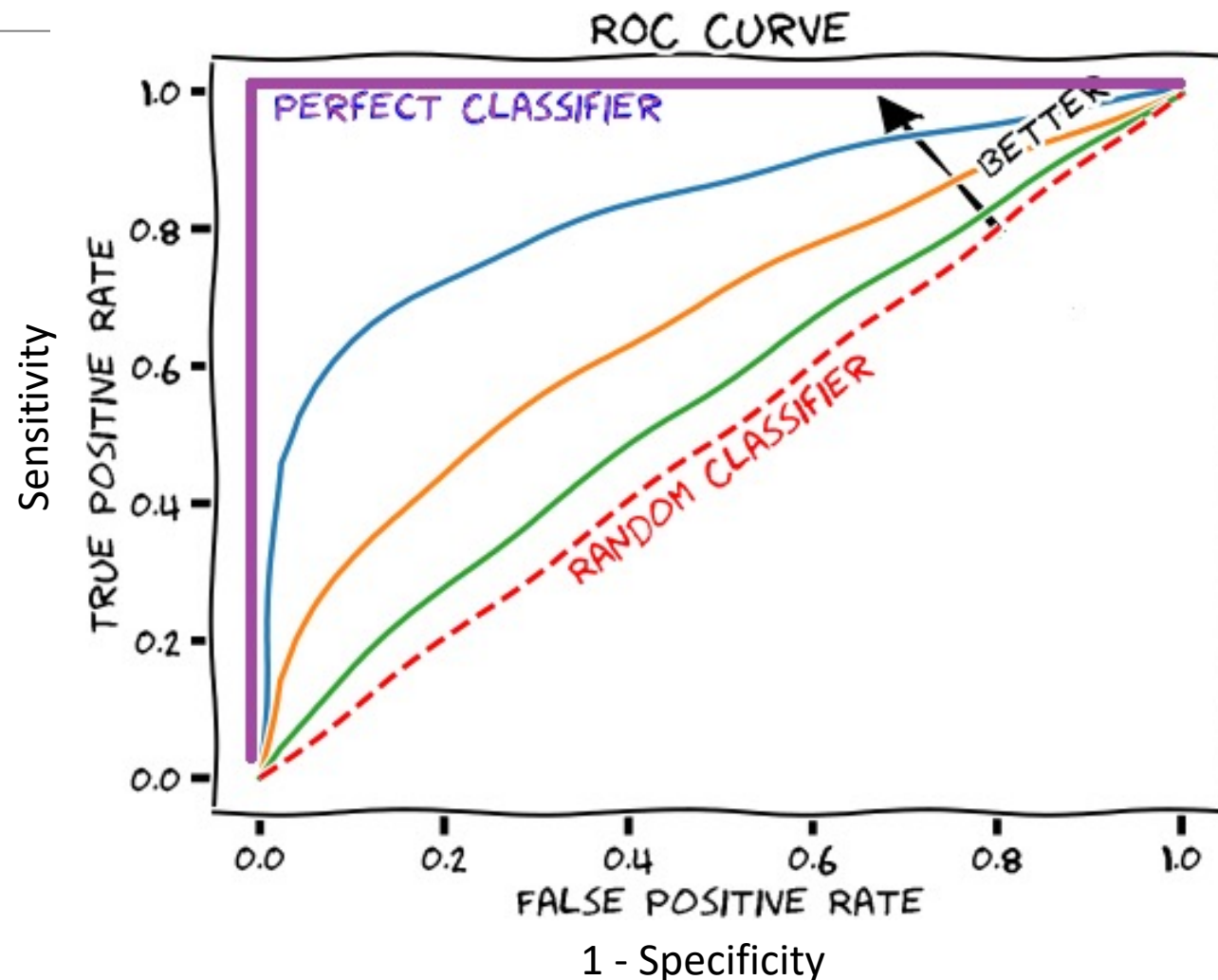
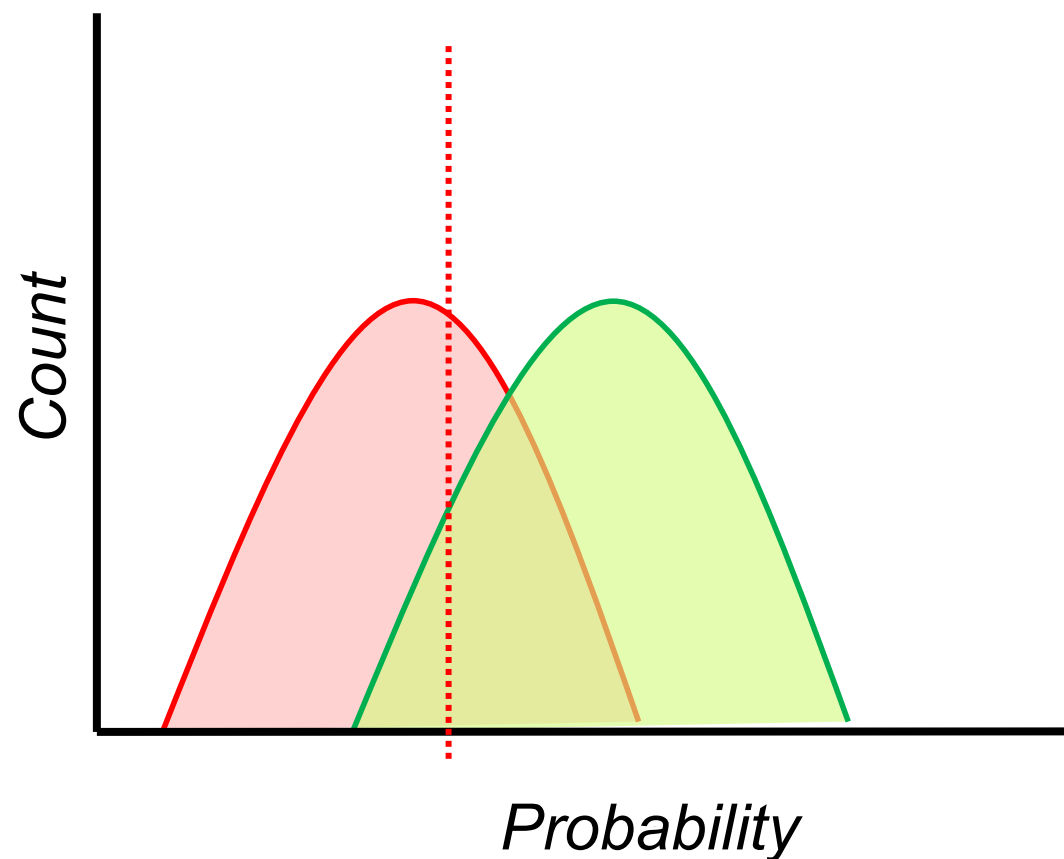
$$FP = C + D$$

$$FN = B$$

$$\text{Sensitivity} = TP / (TP + FN)$$

$$\text{Specificity} = TN / (TN + FP)$$

ROC and AUC



The AUC (area under the curve) is the probability that classifier will rank a random positive sample higher than a random negative sample



2025 SUMMER RESEARCH

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